Axolabs GmbH

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1 Related Documents

This instruction describes the execution of a PNA-HPLC-Assay based on the following documents:

- KU-RGL-SOP-002 (formerly: REG-102) – Methoden-Validierung im GLP/GCP-Bereich (*Method Validation in GLP/GCP Area*) with the following subordinate Instructions, Forms, Spreadsheet Applications:
  - KU-RGL-ESS-002 (formerly: REG-102-S02) – Method Validation Data Report
  - KU-RGL-ESS-005 (formerly: REG-102-S05) – Method Validation Protocol PNA Assay - Liquid Matrices - OnePot
- KU-RGL-SOP-009 (formerly: REG-109) – Durchführung des PNA-HPLC-Assay (*Execution of the PNA-HPLC-Assay*) with the following subordinate Instructions, Forms, Spreadsheet Applications:
  - KU-RGL-ESS-013 (formely: REG-109-S03) – PNA Assay Instruction and Protocol – Liquid Matrices - OnePot
  - KU-RGL-ESS-014 (formerly: REG-109-S04) – Preparation of HPLC Buffers & Stock Solutions
  - KU-RGL-INS-005 (formerly: REG-109-I01) – PNA-HPLC-Assay SST
- KU-RGL-SOP-010 (formerly: REG-110) – Herstellung von Standardlösungen mit einer Präzisionswaage (Preparation of Standard Solutions Using a Precision Balance)
- Development protocol in the project folder AN400.0-24 under the following link:

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The above mentioned documents describe the assay execution under nonGLP-relevant conditions. The parameters and results described in this document are generated from a nonGLP method development, which was conducted following the above mentioned SOPs.

2 Definitions and Abbreviations

°C	Grad Celsius
AB0X	Analytical batch AN400.0-23_02_AB0X
ACN	Acetonitrile
AEX-HPLC	Anion-Exchange High-Performance Liquid Chromatography
CSF	Cerebrospinal fluid
CV	Coefficient of Variation
DNA	Deoxyribonucleic Acid
ed	Edition
EDTA	Ethylenediaminetetraacetic Acid
FDA	Food and Drug Administration
GLP	Good Laboratory Practice
HPLC	High Performance Liquid Chromatography
ICH	International Conference on Harmonization
IFU	Instructions For Use
LC/MS	Liquid Chromatography / Mass Spectrometry
LLOQ	Lower Limit of Quantitation
min	Minute
n.a.; N/A; N.A.	Not applicable
PBS	Phosphate Buffered Saline
PI	Principal Investigator
PNA	Peptide Nucleic Acid
QC	Quality Control
rcf	Relative centrifugal force
RT	Room temperature
SDS	Sodium Dodecyl Sulfate
siRNA	Small Interfering Ribonucleid Acid
SOP	Standard Operating Procedure
SD	Standard Deviation
SST	System Suitability Test(ing)
TCEP	Tris(2-carboxyethyl) phosphine hydrochloride solution
ULOQ	Upper Limit of Quantitation

WS	Working solution

3 ACTIVITIES TO BE PERFORMED

This method development report describes an anion exchange (AEX)-HPLC method with fluorescence detection that allows the specific detection of the analyte 2439-exNA from CSF. 2439-exNA is a dimer composed of two identical siRNAs, which are linked at their sense strands with a tetraethylene glycol linker, with the sequence depicted in Figure 1:

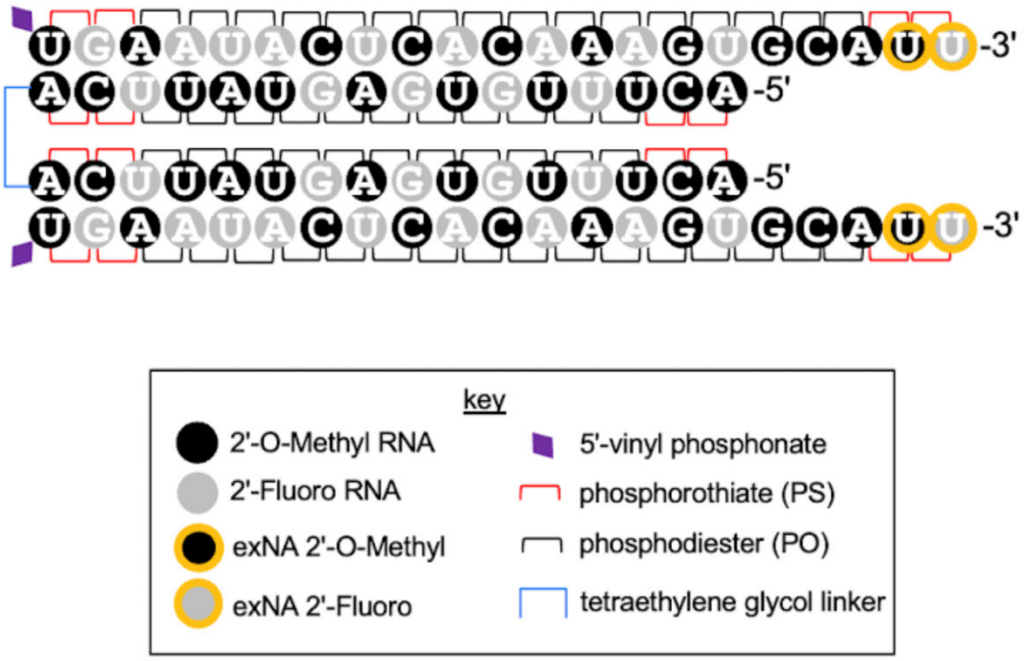

Figure 1 Structure and sequence of the analyte 2439-exNA

The assay is based on the specific hybridization of the antisense strand of 2439-exNA to the complementary PNA-probe. The PNA-probe is N-terminally conjugated with an Atto425-fluorescent dye. The utilized PNA-BroadInst-01 is complementary to the antisense strand of 2439-exNA and has the following sequence:

N-term Atto425-OO-cdc ttt gtg dgt d C-term

The method was developed using artificial CSF as surrogate matrix. Rat or dog CSF was not used as blank matrix in the method development, because it is a rare matrix and difficult to purchase. This procedure is in accordance with the ICH M10 guideline and the 3Rs Principle (Replacement, Reduction and Refinement) intended to reduce animal experiments to an absolute minimum. Nevertheless, dog and rat CSF will be used in the method validation for the selectivity and matrix effect.

This method was developed without having any metabolites of 2439-exNA.

Figure 2 and Figure 3 show the HPLC-profile of 2439-exNA in artificial CSF.

3.1 Summary

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4 Results from Qualification runs in CSF

In this section, the raw data for the determination of calibration curve accuracy (Table 3), confirmation of LLOQ (Table 4), selectivity (Table 5), carry-over (Table 6), accuracy and precision (Table 7), freeze and thaw and short term stability (Table 8) and dilution integrity (Table 9) from 2439-exNA in artificial CSF are shown. The results were generated in two qualifying runs. Calibration standard and QC samples were prepared by spiking 2439-exNA into artificial CSF. Samples were prepared according to the assay procedure described in Section 5.

Two qualifying runs were perfomed (AB03 and AB04). Due to a suspected contamination which lead to signals in the unspiked selectivity samples and one OC-Low stability samples, these tests were prepared freshly and repeated in AB04.

During the qualification runs, the calibration range was between 1.0 ng/mL and 1,000.0 ng/mL. All calibration standards were within ±15 % (±20 % at the LLOQ) of the nominal concentration.

Selectivity and Sensitivity

The LLOQ was confirmed at 1.0 ng/mL in CSF. No signal was detected in corresponding blank CSF samples and the 2439-exNA peaks at the LLOQ of 1.0 ng/mL were at least 5 times the background signal in the analysis run. At the LLOQ, the mean back-calculated concentration was 100.0 % of nominal with a CV of 0.0 %.

The selectivity of the method was demonstrated by comparison of unspiked samples of CSF with the same CSF samples spiked with 2439-exNA at a concentration of the LLOQ. The signal for 2439-exNA in the spiked samples gave an at least 5-fold higher response (peak area) compared to the blank samples.

Carry Over

Carry over CSF samples were run after the 1,000.0 ng/mL calibration standards. There were no distinct analyte signal detectable in the carry-over samples after the 1,000.0 ng/mL standards.

Accuracy and Precision

QC samples were prepared at 4 concentration levels which are covering the calibration curve range:

- at the LLOQ (QC-LLOQ): 1.0 ng/mL
- at 3 times above the LLOQ (QC-Low): 3.0 ng/mL
- at around 50 % of the calibration curve range (QC-Mid): 450.0 ng/mL
- at least at 75 % of the upper calibration curve range (QC-High): 900.0 ng/mL

The QC samples for the determination of the accuracy and precision of the method met the acceptance criteria with following results:

- 100 % of the QC samples at all concentration levels were within ±15 % (±20 % at the LLOQ) of their nominal value.
- The mean accuracies in **AB03** at the 4 concentration levels were 106.7 % (QC-LLOQ), 97.8 % (QC-Low), 102.0 % (QC-Mid) and 98.7 % (QC-High).
- The mean accuracies in **AB04** at the 4 concentration levels were 100.0 % (QC-LLOQ), 96.7 % (QC-Low), 98.2 % (QC-Mid) and 98.4 % (QC-High).
- The mean precisions in **AB03** at the 4 concentration levels were 10.8 % (QC-LLOQ), 2.0 % (QC-Low), 0.3 % (QC-Mid) and 2.1 % (QC-High).
- The mean precisions in **AB04** at the 4 concentration levels were 0.0 % (QC-LLOQ), 0.0 % (QC-Low), 2.9 % (QC-Mid) and 0.2 % (QC-High).

Freeze-Thaw and Short-Term-Stability

The freeze and thaw (one cycle) and short term stability of 2439-exNA in CSF was successfully tested for up to 4 hours. Aliquots of the 2439-exNA at the 2 QC concentration levels QC-Low and QC-High were stored at room temperature for 4 hours and analyzed in triplicate.

After 4 hours at room temperature the mean back-calculated concentrations were at 115.6 % (QC-Low) and 93.1 % (QC-High) of the nominal value.

The QC-Low samples were repeated in AB04: After 4 hours at room temperature the mean back-calculated concentrations were at 91.1 % (of the nominal value

Dilution Integrity

The precision and accuracy of CSF sample dilution was evaluated using a DI-QC standard prepared at concentration levels 500-fold (500,000.0 ng/mL) above the ULOQ of 1,000.0 ng/mL. This QC sample was analyzed and quantified after 1:1,000 dilution in 3 replicates. Samples were diluted with artificial CSF/1 % Tween/ 1 mM TCEP (see section 5.4.3.1).

- The mean accuracy was 93.9 % for the 500,000.0 ng/mL dilution standard (1:1,000).
- The CV was 2.2 % for the 500,000.0 ng/mL dilution standard (1:1,000).

Table 3 Calibration Curve Accuracy in CSF

2439-exNA		Cal-Level-2		Cal-Level-3		Cal-Level-4		Cal-Level-5		Cal-Level-6		Cal-Level-7	
		1.0 ng/mL		5.0 ng/mL		50.0 ng/mL		100.0 ng/mL		500.0 ng/mL		1,000.0 ng/mL	
Run ID	Sample No.	Calc. 2439-ex NA Conc. ng/mL	Acc. criterion +/-20 % of Nominal	Calc. 2439-ex NA Conc. ng/mL	Acc. criterion +/-15 % of Nominal	Calc. 2439-ex NA Conc. ng/mL	Acc. criterion +/-15 % of Nominal	Calc. 2439-ex NA Conc. ng/mL	Acc. criterion +/-15 % of Nominal	Calc. 2439-ex NA Conc. ng/mL	Acc. criterion +/-15 % of Nominal	Calc. 2439-ex NA Conc. ng/mL	Acc. criterion +/-15 % of Nominal
AB03	5 - 10	1.0	100.0%	4.8	96.0%	50.5	101.0%	101.7	101.7%	508.2	101.6%	997.9	99.8%
AB03	41 - 46	1.0	100.0%	4.9	98.0%	50.7	101.4%	101.0	101.0%	503.3	100.7%	987.0	98.7%
Mean		1.00	100.0%	4.85	97.0%	50.60	101.2%	101.35	101.4%	505.75	101.2%	992.45	99.2%
SD (+/-)		0.00		0.07		0.14		0.49		3.46		7.71	
CV (%)		0.0%		1.5%		0.3%		0.5%		0.7%		0.8%	
N		2		2		2		2		2		2	
AB04	4 - 9	1.0	100.0%	4.9	98.0%	49.9	99.8%	101.1	101.1%	501.7	100.3%	996.0	99.6%
AB04	32 - 37	1.0	100.0%	4.9	98.0%	51.0	102.0%	102.1	102.1%	499.7	99.9%	998.6	99.9%
Mean		1.00	100.0%	4.90	98.0%	50.45	100.9%	101.60	101.6%	500.70	100.1%	997.30	99.7%
SD (+/-)		0.00		0.00		0.78		0.71		1.41		1.84	
CV (%)		0.0%		0.0%		1.5%		0.7%		0.3%		0.2%	
N		2		2		2		2				2	

Table 4 Confirmation of LLOQ in CSF

2439-exNA		Cal-Level-1			Cal-Level-2		
		0.0 ng/mL			1.0 ng/mL		
Run ID	Sample No.	Peak Area [mV*min]	Calculated 2439-exNA Conc. ng/mL	Acc. Criterion ≤ 20 % of LLOQ	Peak Area [mV*min]	Calculated 2439-exNA Conc. ng/mL	Acc. Criterion +/-20 % of LLOQ
AN400.0-24_02_AB03	4, 5	0.0000	BQL	0.0%	0.0409	1.0	100.0%
AN400.0-24_02_AB03	40, 41	0.0000	BQL	0.0%	0.0383	1.0	100.0%
AN400.0-24_02_AB04	3, 4	0.0000	BQL	0.0%	0.0366	1.0	100.0%
AN400.0-24_02_AB04	31, 32	0.0000	BQL	0.0%	0.0364	1.0	100.0%
Mean		0.00000	n.a.	0.0%	0.03805	1.00	100.0%
SD (+/-)		0.00000	n.a.		0.00208	0.00	
CV (%)		n.a.	n.a.		5.5%	0.0%	
N		4	4		4	4	

Table 5 Selectivity in CSF

2439-exNA					Blank		QC-LLOQ	
					0.0 ng/mL		1.0 ng/mL	
Run ID	Date	Sex	Lot #	Sample No.	Peak Area [mV*min]	Acc. Criterion ≤ 20 % of LLOQ	Peak Area [mV*min]	Calculated 2439-exNA Conc. ng/mL
AB04	19.Aug.2024	n.a.	240719	20, 23	0.0000	0.0%	0.0366	1.0
	19.Aug.2024	n.a.	240719	21, 24	0.0000	0.0%	0.0365	1.0
	19.Aug.2024	n.a.	SYN154	22, 25	0.0000	0.0%	0.0368	1.0

Table 6 Carry-over in CSF

2439-exNA		Cal-Level-2			Cal-Level-7			Carry Over		
		1.0 ng/mL			1,000.0 ng/mL			0.0 ng/mL		
Run ID	Sample No.	Peak Area [mv*min]	Calc. 2439-exNA Conc. ng/mL	Acc. Criterion +/-20 % of Nominal	Peak Area [mv*min]	Calc. 2439-exNA Conc. ng/mL	Acc. Criterion +/-15 % of Nominal	Peak Area [mv*min]	Calc. 2439-exNA Conc. ng/mL	Acc. Criterion ≤ 20 % of LLOQ
AB03	5, 10, 11	0.0409	1.0	100.0%	43.2717	997.9	99.8%	0.0000	BQL	0.0%
AB03	41, 46, 47	0.0383	1.0	100.0%	42.7999	987.0	98.7%	0.0000	BQL	0.0%
AB04	4, 9, 10	0.0366	1.0	100.0%	42.6621	996.0	99.6%	0.0000	BQL	0.0%
AB04	32, 37, 38	0.0364	1.0	100.0%	42.7738	998.6	99.9%	0.0000	BQL	0.0%

Table 7 Accuracy and Precision in CSF

2439-exNA		QC-LLOQ		QC-Low		QC-Mid		QC-High	
		1.0 ng/mL		3.0 ng/mL		450.0 ng/mL		900.0 ng/mL	
Run ID	**Sample No.**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion +/-20 % of Nominal**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion +/-15 % of Nominal**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion +/-15 % of Nominal**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion +/-15 % of Nominal**
AB03	23, 26, 29, 32	1.0	100.0%	3.0	100.0%	460.3	102.3%	909.0	101.0%
	24, 27, 30, 33	1.0	100.0%	2.9	96.7%	458.2	101.8%	872.9	97.0%
	25, 28, 31, 34	1.2	120.0%	2.9	96.7%	458.3	101.8%	883.3	98.1%
Mean		1.07	106.7%	2.93	97.8%	458.93	102.0%	888.40	98.7%
SD (+/-)		0.12		0.06		1.18		18.58	
CV (%)		10.8%		2.0%		0.3%		2.1%	
N		3		3		3		3	
AB04	23, 12, 13, 14	1.0	100.0%	2.9	96.7%	432.8	96.2%	884.1	98.2%
	24, 27, 28, 29	1.0	100.0%	2.9	96.7%	450.9	100.2%	886.2	98.5%
	25	1.0	100.0%	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Mean		1.00	100.0%	2.90	96.7%	441.85	98.2%	885.15	98.4%
SD (+/-)		0.00		0.00		12.80		1.48	
CV (%)		0.0%		0.0%		2.9%		0.2%	
N		3		2		2		2	

Table 8 Freeze and thaw and short term stability in CSF

2439-exNA				QC-Low 3.0 ng/mL		QC-High 900.0 ng/mL	
Run ID	**Date**	**Sample No.**	**Time Point**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion +/-15 % of Nominal**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion +/-15 % of Nominal**
AN400.0-24_02_AB03	14.Aug.2024	13, 16	1 Cycle 4 hours at RT	3.2	106.7%	844.9	93.9%
	14.Aug.2024	14, 17		4.2	140.0%	828.1	92.0%
	14.Aug.2024	15, 18		3.0	100.0%	839.5	93.3%
Mean				3.47	115.6%	837.50	93.1%
SD (+/-)				0.64		8.58	
CV (%)				18.5%		1.0%	
N				3		3	
AN400.0-24_02_AB04	19.Aug.2024	16	1 Cycle 4 hours at RT	2.8	93.3%	n.a.	n.a.
	19.Aug.2024	17		2.5	83.3%	n.a.	n.a.
	19.Aug.2024	18		2.9	96.7%	n.a.	n.a.
Mean				2.73	91.1%	n.a.	n.a.
SD (+/-)				0.21		n.a.	
CV (%)				7.6%		n.a.	
N				3		0	

Underlined values failed acceptance criteria, but were used to calculate Mean, SD and CV.
QC-Low failed the acceptance criteria for Mean and CV in AB03. The QC-Low stability samples were prepared freshly and was repeated in AB04.

Table 9 Dilution Integrity in CSF

2439-exNA		Dilution 1:1,000	
		500,000.0 ng/mL	
Run ID	**Sample No.**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion +/-15 % of Nominal**
AN400.0-24_02_AB03	36	457,808.0	91.6%
	37	475,644.4	95.1%
	38	475,285.2	95.1%
Mean		469,579.20	93.9%
SD (+/-)		10,195.74	
CV (%)		2.2%	
N		3	

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[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

6 Records

The following records in addition to the above mentioned Forms, Master or Spreadsheet Applications are required in this process:

- N/A

7 Change History

Edition, Section(s)	Change	Rationale
Ed 01	New document	Assay description after the method development